

Ahmed Fathy

Software Engineer

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PROFESSIONAL SUMMARY

Hardware and Embedded Systems Engineer with hands-on experience in FPGA development, digital circuit design, and ASIC flow. Skilled in RTL design using Verilog/VHDL, and experienced in validating complex architectures including pipelined processors and cryptographic accelerators.

TECHNICAL SKILLS

- **Digital Design & Verification:** Verilog HDL, VHDL, RTL Design, RISC-V ISA, MIPS Architecture, Pipeline Design, RTL-to-GDSII Flow, OpenLane2, DRC/LVS Verification, Static Timing Analysis, FSM Design
- **Development Tools & Platforms:** ModelSim, QuestaSim, Intel Quartus Prime, GTKWave, ESP32, AVR/Arduino
- **Programming Languages:** C/C++ (Advanced), Python, x86 Assembly, MIPS Assembly

EDUCATION

- **Cairo University, Faculty of Engineering** Giza, Egypt
B.Sc. in Computer Engineering; GPA: 3.6 2022

PROJECTS

- **5-Stage Pipelined Processor (RISC)** | *VHDL, QuestaSim, Computer Architecture* Source Code | 2025
 - **Core Architecture:** Engineered a 32-bit 5-stage pipelined processor in VHDL, supporting a custom ISA of over 25 instructions.
 - **Hazard Resolution:** Optimized throughput by implementing Hazard Detection and Data Forwarding units to resolve control and data dependencies.
 - **Verification:** Validated functional correctness through rigorous simulation and testbench environments in QuestaSim.
- **CNN Convolution Accelerator** | *Verilog, OpenLane2, ASIC Design* Source Code | 2025
 - **Architecture:** Designed a synthesizable 2D convolution accelerator using Systolic Arrays and fully pipelined dataflow to maximize MAC throughput.
 - **Memory Hierarchy:** Implemented DMA-based loading and dual-port SRAM buffering to support up to 16x16 kernel execution.
 - **Physical Design:** Executed RTL-to-GDSII synthesis using OpenLane2, validating physical feasibility and achieving timing closure.
- **AES Encryption/Decryption Engine** | *Verilog, FPGA, Cryptography* Source Code | 2023
 - **Core Logic:** Synthesized robust AES cryptographic core in Verilog HDL supporting 128-bit, 192-bit, and 256-bit key standards.
 - **Datapath:** Engineered hardware-optimized modules for SubBytes, ShiftRows, MixColumns, and AddRoundKey transformations.
 - **Validation:** Verified cryptographic correctness against standard NIST test vectors using rigorous debug tracing.
- **Tactic (Robotics & AI)** | *Embedded Systems* Source Code | 2024
 - **System Integration:** Developed browser-controlled robotics suite integrating AI computer vision with low-latency hardware control.
 - **Firmware:** Programmed ESP32 microcontrollers to handle WebSocket communication for real-time motor control and telemetry.
 - **Backend Processing:** Deployed YOLOv5 and OpenCV on host server to process visual feedback and automate robotic decision-making.
- **Tactic (Robotics & AI)** | *Real-Time Systems, WebSockets, Embedded* Source Code | 2024
 - **Real-Time Messaging System:** Engineered a low-latency WebSocket communication layer to integrate a web dashboard with embedded hardware.
 - **Reliability:** Ensured reliable message delivery and system synchronization between the browser interface and the robotic unit.

EXPERIENCE

- **Al Nada Scientific Office** | *Software Developer (Part-time)* *Cairo, Egypt | June 2022 – 2024*
 - **Full-Stack Development:** Designed and built a custom ERP application using React.js, Express.js, and PostgreSQL to replace legacy spreadsheet-based workflows.
 - **Web Development:** Developed and enhanced company's official website using Next.js, optimized SEO for Google search rankings, and integrated ecommerce features for online product sales.
 - **Desktop Deployment:** Packaged the ERP application with Electron for local server deployment, enabling reliable offline operation.

STUDENT ACTIVITIES

- **IEEE** | *UI/UX Designer - Frontend Team Leader* *Cairo | Oct 2025 – Present*
 - **Portfolio Platform Development:** Engineered comprehensive web platform featuring workshop and event management systems with admin dashboard for content creation, role-based access control enabling instructors to upload course materials and students to register for workshops and events.
- **Technical Center for Career Development (TCCD)** | *Frontend Developer* *Cairo | June 2025 – Present*
 - **Portfolio Platform Development:** Developed comprehensive platform with event pages, admin dashboard for event creation, role-based access for instructors and students, location management, event galleries, and statistical analysis pages.
 - **HR Management & Judging System:** Built HR system managing team attendance with custom form builder and reviewer functionality similar to Google Forms, alongside event judging system for evaluating and assessing teams applying to events.
- **Physics Helping District (PhD)** | *Team Leader* *Cairo | June 2022 – 2024*
 - **Mentorship:** Led peer tutoring initiative in electronics, achieving measurable academic improvement.
- **NASA Space Apps Challenge** | *Participant – Project: "Orrery"* *Cairo | Oct 2024*
 - **Orrery App:** Developed an interactive orrery app visualizing near-Earth objects using React and Three.js.